BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: 1465 / 1466

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — بیالوجی (معروضی) — گروپ پہلا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1465

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Five kingdom classification system was introduced by ___

(A) (B)

(C) (D)

Q2. The function of chloroplast is ___

(A) (B)

(C) (D)

Q3. Chiasmata is formed during ___

(A) (B)

(C) (D)

Q4. ATP was discovered by ___

(A) (B)

(C) (D)

Q5. Which acid converts the inactive enzyme pepsinogen into its active form?

(A) (B)

(C) (D)

Q6. Which protein is blood clotting?

(A) (B)

(C) (D)

Q7. "Scientific knowledge is the common heritage of mankind" is saying of ___

(A) (B)

(C) (D)

Q8. Entomology is the study of which organisms?

(A) (B)

(C) (D)

Q9. The logical consequences of hypotheses are ___

(A) (B)

(C) (D)

Q10. Which vitamins act as Coenzymes?

(A) (B)

(C) (D)

Q11. The formation of new tumors is known as ___

(A) (B)

(C) (D)

Q12. In which organelle ribosomes are constructed?

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — بیالوجی (انشائیہ) — حصہ اول — گروپ پہلا

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Define physiology and taxonomy. (ii) What is horticulture and how is it related to agriculture. (iii) Differentiate between micromolecules and macromolecules with examples. (iv) Define biological method (v) Differentiate between law and theory. (vi) Define bioinformatics. (vii) What is fluid mosaic model. (viii) What is exocytosis? Give its importance.

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) What are aims of classification (ii) What are simple multicellular organisms. (iii) What is phragmoplast and how it is formed. (iv) What are effects of Non-disjunction. (v) What are blebs. (vi) Define Mitosis (vii) Define photosystem. (viii) Write down balanced equation of photosynthesis

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Differentiate between Anabolism and Catabolism. (ii) Define metabolic pathways. (iii) What is meant by Enzyme Denaturation, How it affects on enzyme action. (iv) What is the effect of Dengue fever on the Platelets (v) Define Assimilation of Food. (vi) Give the Role of Gastrin Hormone in Digestion. (vii) What is "Source" Give example. (viii) What is Megakaryocytes.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. Why medicine and farming is important as career in biology. 2+2

b. Define muscle tissues. Explain its three types. 2+3

Q6. 6.a. Justify why viruses are excluded from five kingdom system. 4

b. Interpret the importance of the fermentation. 3+2

Q7. 7.a. What is oral cavity? Describe the role of oral cavity in digestion of food. 1+3

b. Explain arterial system of man. 5

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **بیالوجی** (**معروضی**) — **گروپ دوسرا**

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1466

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Point of attachment of non-sister chromatids of homologous chromosomes is called.

(A) (B)

(C) (D)

Q2. Genetic material in nucleus is in loose thread like form called ___

(A) (B)

(C) (D)

Q3. Who used the term 'enzyme' first time.

(A) (B)

(C) (D)

Q4. Which one is main photosynthetic pigment.

(A) (B)

(C) (D)

Q5. Where are villi found

(A) (B)

(C) (D)

Q6. The normal heart rate in man is ___ times per minute

(A) (B)

(C) (D)

Q7. Which is the latest profession in field of Biology.

(A) (B)

(C) (D)

Q8. Jabir Bin Hayan was born in ___

(A) (B)

(C) (D)

Q9. The information such as names, dates and values made from observations and experiments is called ___

(A) (B)

(C) (D)

Q10. Which one organism is simple multicellular

(A) (B)

(C) (D)

Q11. Ribosomes are chemically composed of equal amount of ribosomal RNA (rRNA) and ___

(A) (B)

(C) (D)

Q12. The shrinkage of cytoplasm is called ___

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **بیالوجی (انشائیہ) — حصہ اول — گروپ دوسرا**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) What are six major Bioelements in Human. (ii) What is Farming. (iii) What is tissue level. Give examples. (iv) What is Science. (v) What is Deduction. (vi) Write the effects on the Human body after bitting Mosquitoes. (vii) Differentiate between magnification and Resolution (viii) Give the similarities between Mitochondria and Chloroplast.

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Write any two importance of biodiversity. (ii) How Fungi differ from plants? Write two points. (iii) What is G1 phase? (iv) What is importance of Apoptosis? (v) Differentiate between somatic cells and germ line cells. (vi) What is difference between diploid and haploid? (vii) Sketch molecular structure of ATP. (viii) Define cellular respiration.

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Which state is called saturation of Active sites? (ii) Who and when proposed induced fit model? (iii) Write use of enzymes in paper industry. (iv) State two functions of vitamin D. (v) Write weight and size of Liver in adult human. (vi) Give functions of neutrophils and basophils. (vii) What are root hairs? and write its function. (viii) Write any two preventive measures to prevent dengue fever.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5(a) Explain popuiation level and community level with examples. 2+2

(b) What are compound tissues? Explain xylem tissue and phloem tissue in detail. 1+2+2

Q6. 6(a) Write a note on causes and effects of deforestation. 2+2

(b) Describe mechanism of respiration. 1+1+3

Q7. 7(a) Describe swallowing and peristalsis action of food. 2+2

(b) How would you relate the internal structure of root with the uptake of water and salts. 3+2

> Note: Q2 scan layout jumbled: left column labels printed (i)(iii)(v)(vii)(viii), right column (ii)(iv)(vi); transcribed in intended (i)-(viii) order. Board typos kept verbatim: 'popuiation', 'bitting', 'Winhelm Kuhne', '40^{b}'.